Peidi Song

Emails: ppppaidy@gmail.com | psong45@gatech.edu
Phone: 404-409-4853 | Address: 7212 Ivy Chase Way NE, Atlanta, GA, 30342

Objective

I am actively looking for an internship position in 2026 Summer. My research interests lie in computer network, computer architecture, and networked system. I like designing efficient and high-performance software applications and hardware architecture. In addition, I am expert in data structure and algorithms. If you find me qualified, please feel free to contact me.

Education

Georgia Institute of Technology | Atlanta, GA, U.S.
August 2024 – May 2029 (expected)
Ph.D. in Computer Science
Advisors: Prof. Ahmed Saeed & Prof. Alexandros Daglis
Georgia Institute of Technology | Atlanta, GA, U.S.
August 2022 – May 2024
Master of Science in Electrical & Computer Engineering, GPA 3.90
Peking University | Beijing, China
September 2018 – July 2022
Bachelor of Science in Computer Science, GPA 3.65

Ongoing Research Project

Resource-Aware Heterogeneous Microservice Request Load Balancing
October 2024 – Present
Georgia Institute of Technology

- Noticed that existing load balancers assume all requests are bottlenecked on a single resource and only react to the overall application performance such as request latency. However, when different types of requests demand different types of resources (e.g., CPU and memory bandwidth), these approaches underutilize the non-bottlenecked resources, leading to throughput loss.
- Proposed a lightweight greedy heuristic to leave headroom for requests demanding non-bottlenecked resources, allowing for balancing and maximizing the utilization of all types of resources on heterogeneous machines.
- Conducted simulation to show that a resource-aware load balancer that utilizes this heuristic has the potential to improve resource utilization by up to 78%, leading to an up to 25% throughput improvement compared to state-of-the-art approaches.
- *Currently working on the engineering implementation and targeting a conference paper.*

Publications

- Size Zheng, Siyuan Chen, **Peidi Song**, Renze Chen, Xiuhong Li, Shengen Yan, Dahua Lin, Jingwen Leng, and Yun Liang. "Chimera: An Analytical Optimizing Framework for Effective Compute-intensive Operators Fusion." In *2023 IEEE International Symposium on High-Performance Computer Architecture (HPCA),* pp. 1113-1126. IEEE, 2023.
- **Peidi Song**, Alexandros Daglis, Michael Filler, and Ahmed Saeed. "Algorithmic Tradeoff Exploration for Component Placement and Wire Routing in Nanomodular Electronics." *arXiv preprint arXiv: 2510.03126* (2025).

Skills

Programming: C, C++, Python, Cuda, MPI, OpenMP, Verilog, HLS, Pascal
Software: DPDK kernel-bypass network stack, Apache Thrift RPC framework, Docker and Kubernetes container management tool for cloud applications
Hardware: CPU (including AVX-512), GPU (including Tensor Cores), NPU, FPGA
Communication: technical reports, team cooperation, presentations, writing
Languages: English (fluent), Chinese (native), Japanese (fluent)

Relevant Courseworks

Datacenter Networks & Systems, Computer Networks, Advanced Computer Architecture, Parallelizing Compilers, High Performance Parallel Computing, Compiler Principles, Design of Digital Logic

Selected Past Research Projects

Algorithmic Tradeoff Exploration for Design Automation in Nanomodular Electronics
August 2023 – May 2024
Georgia Institute of Technology

- Nanomodular electronics (NE) fabricate high-performance nanomodular components in distributed fabs and assemble them with a 3D-printer-like machine. This allows NE to manufacture chips on demand with low capital cost, but raises new challenges in design automation step due to imprecise component deposition.
- Formulated the component placement and wire routing problem for NE, identifying unique constraints and the tradeoff space between time-to-solution and solution quality.
- Presented NE design automation workflow and adapt algorithm that capture design space tradeoffs and balance all requirements.
- Conducted an evaluation demonstrating how our suite of algorithms enables selective movement in the tradeoff space, based on the target optimization metric. For example, we show an opportunity for 108x improvement in end-to-end manufacturing time at the cost of 21% increase in the target circuit's total wire length.
- *Published an arXiv paper.*

An Analytical Optimizing Framework for Effective Compute-intensive Operator Fusion
October 2021 – October 2022
Center for Energy-efficient Computing and Applications, Peking University

- Figured out that existing machine learning compilers lack both precise analysis and efficient optimization for compute-intensive operator chains on different accelerators.
- Proposed an optimizing framework to efficiently improve the locality of compute-intensive operator chains on different hardware accelerators.
- Achieved up to 2.29x, 1.64x and 1.14x speedups compared to state-of-the-art compilers for CPU, GPU and NPU respectively.
- *Published a conference paper in HPCA 2023.*

Modeling and Optimization for Convolution Operator towards Ascend NPU Platform
January 2022 – June 2022
Center for Energy-efficient Computing and Applications, Peking University

- Indicated existing Tensor operator optimization works on Ascend NPU lacked a performance model that helped developers to reason about expected performance. They did not fully utilize hardware resources, either.
- Presented an abstraction of hardware architecture and convolution program loop nests, as well as the general performance cost model on Ascend NPU.

- Conducted experiments to show the accuracy of the cost model achieves 95.0% rank accuracy and 0.9861 R-squared. With the performance model, the optimized convolution operator program achieves on average 4.76x speedup to current works.

Teaching Assistant Experience

Data Structure and Algorithm (B)
Spring 2021
School of Electrical Engineering and Computer Science, Peking University
This is a course for departments other than EECS. It covers fundamental data structures such as list, binary tree, tree and forest, and graph; sorting and searching; data abstraction and problem solving.

- Helped students with their labs and graded their labs every week.
- Involved in designing and grading final exam. No one failed the course in this section with our help.